

“Trash To Treasure”

A report submitted for the course of

Application Development_Full Stack Development Using AI **II B.Tech I Semester**

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DATA SCIENCE

CERTIFICATE

This is to certify that this bonafide record of the Application Development entitled “**Trash 2 Treasure**” submitted by **Mr. Vishnu Konda (2411CS030183)**, **Ms. Mounika Kota (2411CS030226)**, **Ms. Akshaya Kulpaguri (2411CS030235)**, **Ms. Nakshatra Lingaraju (2411CS030249)** of **II year I semester** to the Malla Reddy University, Hyderabad. This Bonafide record of work carried out by us under the guidance of our supervision. The contents of this report, in full or in parts, have not been submitted to any other organization for the award of any Degree.

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DECLARATION

We hereby declare that the project report entitled “Trash 2 Treasure” has been carried out by us. This work has been submitted to the Department of Computer Science and Engineering (Data Science), Malla Reddy University, Hyderabad in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology. We further declare that this project work has not been submitted in full or part for the award of any other degree in other educational institutions.

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ABSTRACT

Trash2Treasure (T2T) is a smart digital platform designed to encourage responsible waste disposal by rewarding users for proper waste management. The main goal of T2T is to transform the way people see waste — not as garbage, but as a valuable resource. Through this system, users can log their waste disposal activities, earn credits for following correct waste segregation methods, and redeem those credits for rewards, discounts, or benefits from partner organizations. The platform focuses on promoting a circular economy where waste is minimized, recycled materials are reused, and individuals are motivated to contribute to a cleaner environment. T2T combines awareness with technology to make waste management easy, engaging, and rewarding. Users can track their contributions, learn best practices for segregation, and become part of a community that values sustainability. Unlike traditional systems that rely on physical smart bins or hardware, T2T is a fully software-based solution that can be accessed via web or mobile devices, making it scalable and affordable. By integrating gamification, digital incentives, and eco-education, T2T aims to bridge the gap between citizens and environmental responsibility. In short, Trash2Treasure turns everyday waste disposal into a rewarding, community-driven step toward a greener and cleaner future.

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Chapter – 1

Introduction

The management of municipal solid waste (MSW) is a critical environmental challenge for urban centers worldwide. Rapid urbanization and shifting consumption patterns have caused an exponential increase in waste generation, with global MSW projected to grow from 2.01 billion to 3.04 billion metric tons by 2050.¹ This surge strains municipal infrastructure, public health, and the environment. Poor waste management, especially the failure to segregate at the source, results in overflowing landfills, pollution, and greenhouse gas emissions.² This crisis is driving a shift from a linear "take-make-dispose" model to a circular economy, which treats waste as a valuable resource, aiming to eliminate waste and keep materials in use. The Trash2Treasure (T2T) platform is a direct response to this imperative, designed to catalyze the transition towards a circular economy at the grassroots level. The core proposition of T2T is the deployment of a fully software-based, hardware-agnostic incentive model that prioritizes user engagement and behavioural change over capital-intensive infrastructure. Unlike conventional "smart city" solutions that often require significant investment in specialized hardware such as sensor-equipped bins, T2T leverages the ubiquitous technology already in the hands of citizens: the smartphone.

By providing an accessible web and mobile platform, T2T empowers individuals to become active participants in the waste management ecosystem. The system converts the routine act of waste disposal into an engaging, gamified experience where responsible actions, such as correct waste segregation, are recognized and rewarded with digital points. These points function as a virtual currency, redeemable for discounts, services, or products from partner organizations.

Chapter – 2

Review of Relevant Literature

Several studies and applications have explored digital and smart waste management systems, focusing on automation, monitoring, and sustainability. Projects such as IoT-based smart bins and reward-based recycling apps like RecycleBank have shown the potential of technology in promoting eco-friendly behavior. However, most existing solutions are hardware-dependent or lack user engagement and AI integration. Recent research also highlights the growing role of artificial intelligence in waste classification and cloud technologies in real-time data management. Building on these insights, the Trash2Treasure (T2T) platform combines **AI-based waste recognition, cloud data storage, and reward-driven user engagement** to create an accessible, scalable, and community-centered solution for smart waste management.

Chapter – 3

Methodology

To promote sustainable waste management, encourage recycling habits, and provide users with an engaging reward-based experience, the **Trash2Treasure (T2T)** application follows a structured methodology integrating user authentication, waste data management, AI-powered classification, reward allocation, and cloud-based storage.

1.1. Problem Identification: The main problem identified is the inefficient handling of household and community waste due to a lack of user awareness, motivation, and digital tracking mechanisms. Traditional waste management systems often fail to engage citizens directly or reward eco-friendly actions.

1.2. Requirement Analysis:

- User Authentication (Sign Up & Login)
- Waste submission with image and category
- Automatic classification of waste using AI models
- Reward points calculation based on type and weight of waste
- User dashboard to track submissions and earned points

1.3. System Design: A modular system design was developed using architecture flow diagrams to represent interactions between users, backend, and the database.

- **Authentication Module:** Manages secure user login and registration.
- **Waste Submission Module:** Enables image upload, type selection, and submission.
- **AI Module:** Classifies waste and suggests category-based points.
- **Reward Management Module:** Calculates and updates user credits.

1.4. Implementation

The system is implemented using:

Frontend: JavaScript

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Modules:

- **Login & Registration:** Secure user authentication using JWT.
- **Waste Upload:** Upload waste images with category and weight input.
- **Dashboard:** Displays reward points, history, and statistics.
- **Admin Panel:** Manages verification and analytics.
- **Notifications:** Sends reward updates and collection alerts.

Testing

Each module was tested individually for:

- Proper authentication and session handling
- Accuracy of AI-based waste classification
- Correct calculation and update of reward points
- Validation of form inputs and image uploads
- Consistent performance across web and mobile platforms

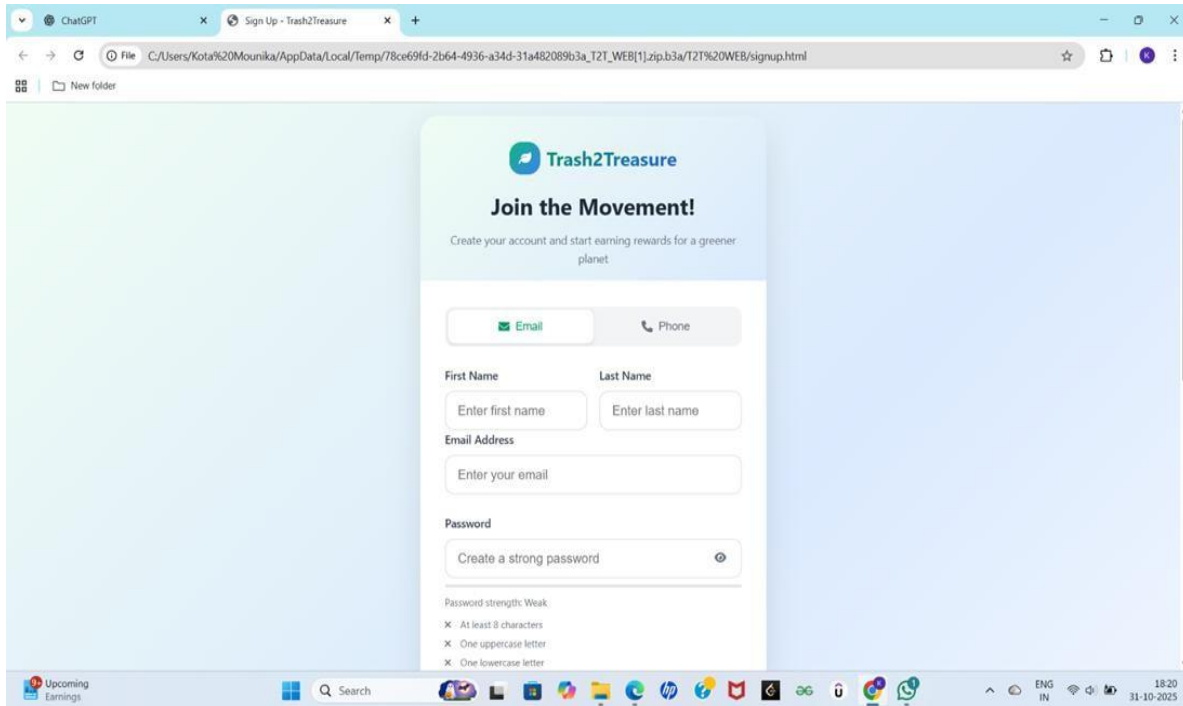
1.3 Deployment: : The complete application is deployed on Render for backend services and Vercel for the web frontend.

MongoDB Atlas hosts the database, and Cloudinary manages media storage.

The mobile application can be deployed through Expo or packaged for Android/iOS.

Chapter – 4

Results



The screenshot shows a web browser window with the address bar displaying a local file path. The page content is a sign-up form for 'Trash2Treasure'. The form is titled 'Join the Movement!' and includes the subtext 'Create your account and start earning rewards for a greener planet'. There are two registration options: 'Email' (highlighted in green) and 'Phone'. Below these are input fields for 'First Name', 'Last Name', 'Email Address', and 'Password'. The 'Password' field has a strength indicator showing 'Weak' and a list of requirements: 'At least 8 characters', 'One uppercase letter', and 'One lowercase letter'. The Windows taskbar is visible at the bottom with various application icons and a system clock showing 18:20 on 31-10-2025.

Fig 1 – Sign up page

The above figure represents the sign-up page of the application. It allows new users to create an account by providing details such as name, email, password, and other required information. Once the data is submitted, it is stored securely in the database for future authentication.

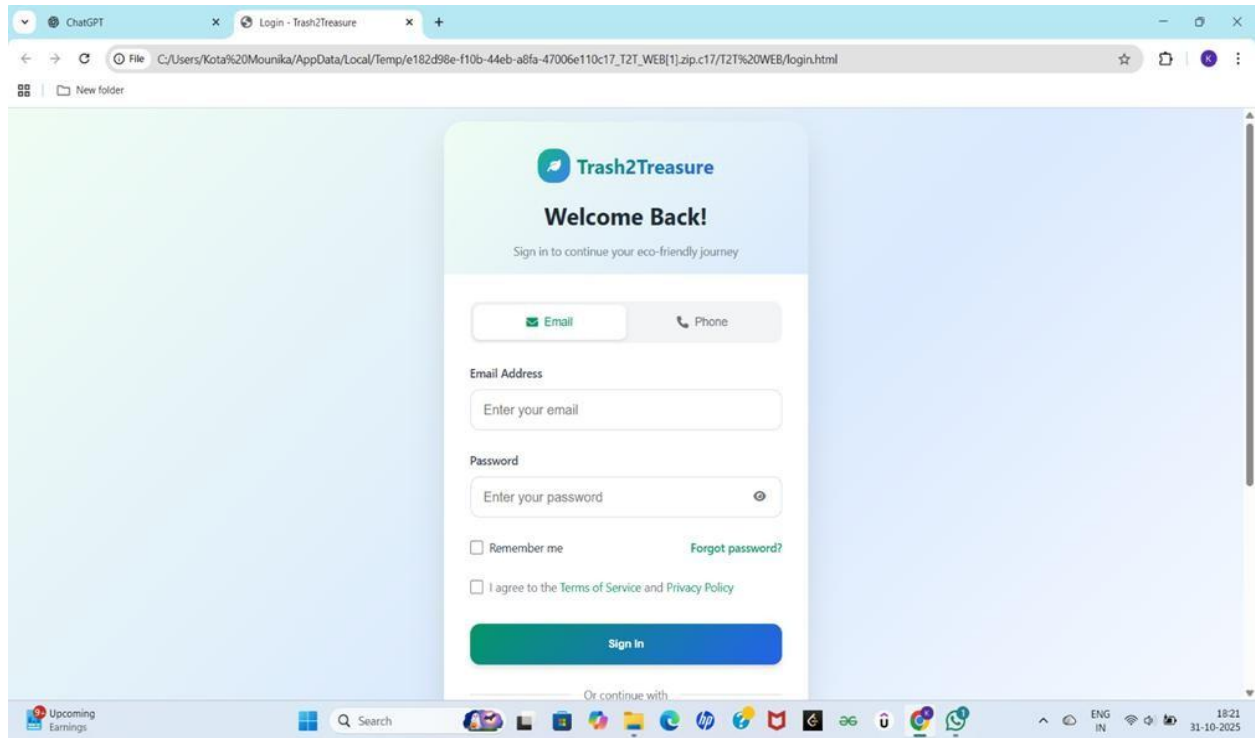


Fig 2 – Login page

The user can log in to the application using the username and password provided during the sign-up process. Upon entering valid credentials, the user gains access to their personalized dashboard and application features.

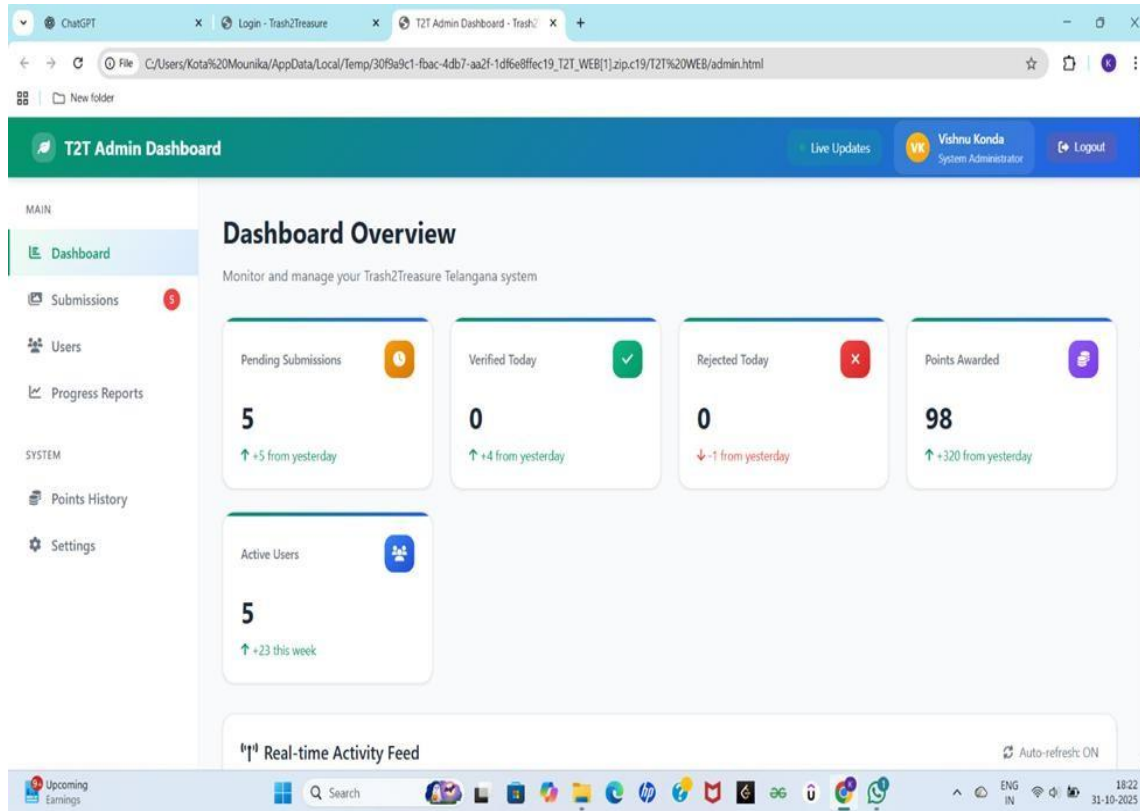


Fig 3 – Dashboard page

It serves as the central interface for users to view and manage all their tasks efficiently. The dashboard displays the status of each task under three main categories — Pending, Ongoing, and Completed. This helps users easily track their progress and prioritize their work.

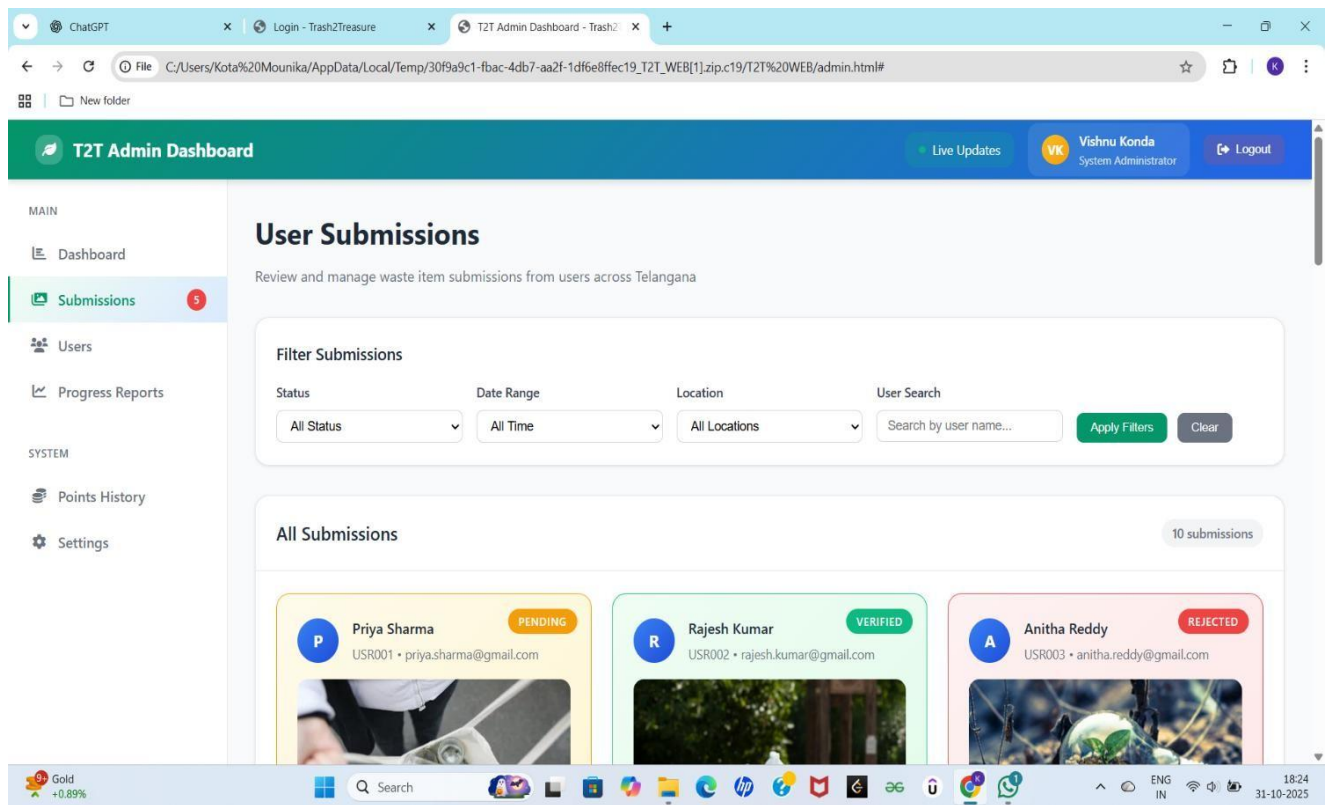


Fig 3.1 - Dashboard page showing Notifications

The above figure represents the notification section within the dashboard. This section provides real-time updates and alerts related to ongoing tasks. Users are notified about task deadlines or reminders.

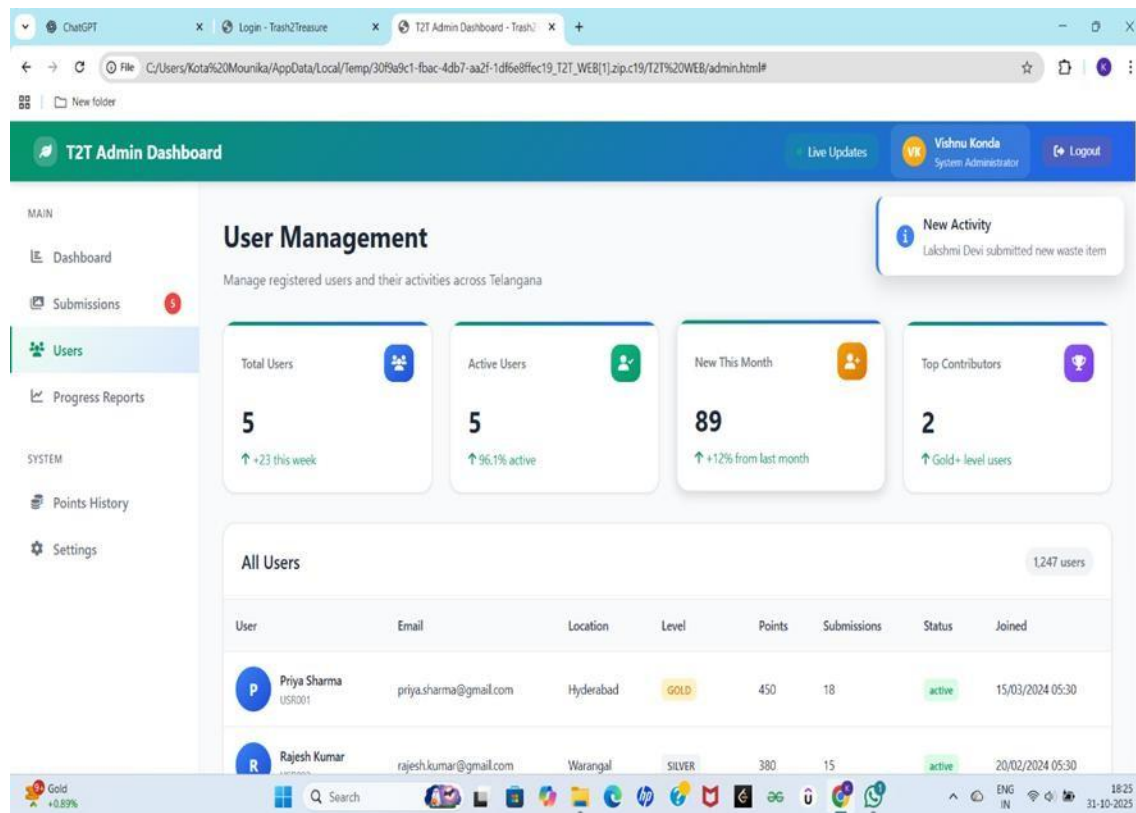


Fig 3.2 - Dashboard page showing Alert

The above figure shows the alert section of the dashboard that highlights the pending tasks. This section helps users stay informed about tasks that are yet to be started or completed. The alert system reminds users about deadlines and encourages them to take timely action to prevent delays in project progress.

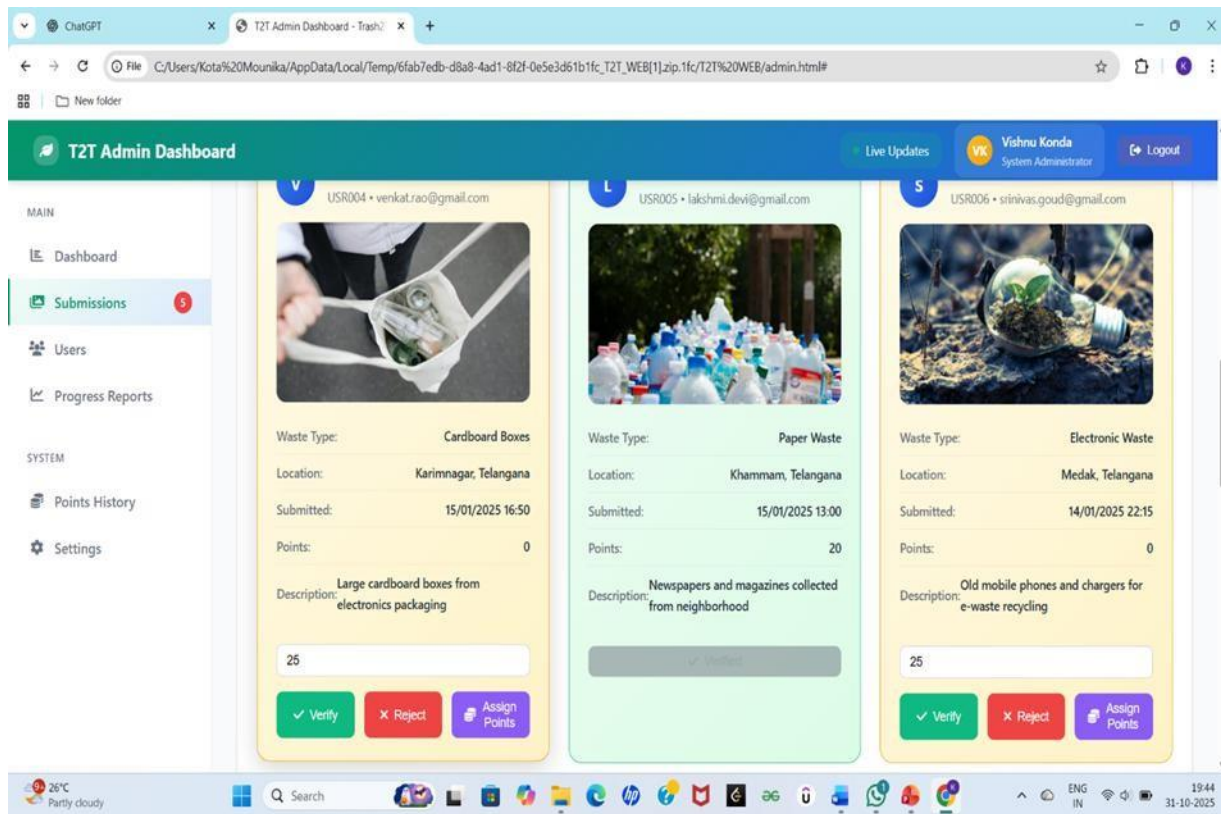


Fig 4 - Create a new task

The above figure represents the Task Creation. This page allows users to create new tasks by entering details such as task name, description, start date, due date, and priority level and category. The created tasks are automatically categorized and displayed on the dashboard under the appropriate status section.

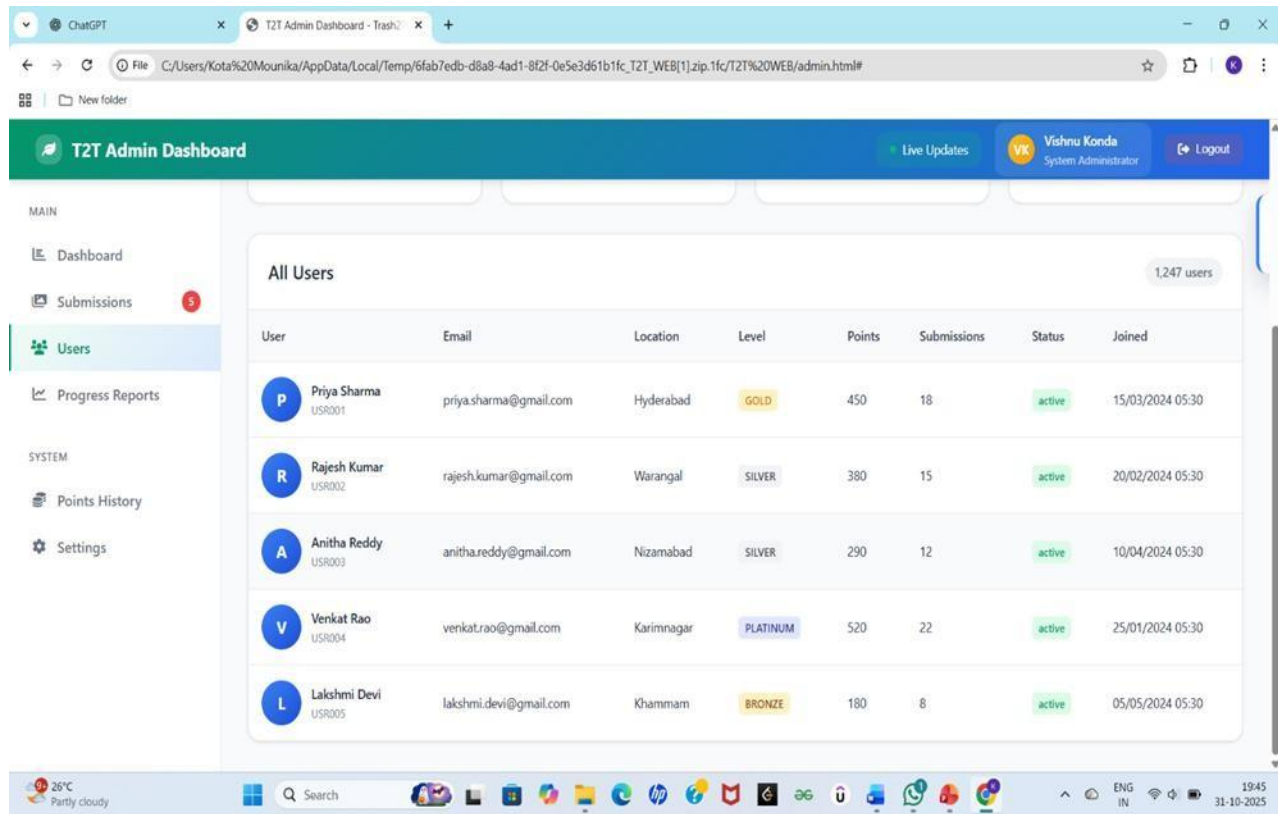


Fig 5 - View/Edit tasks

This version of the dashboard provides users with a complete overview of all tasks along with their detailed information. Each task can be viewed or edited directly from this interface. Additionally, tasks are categorized based on priority levels, such as Personal and Professional, allowing users to efficiently manage both work and personal responsibilities within the same application.

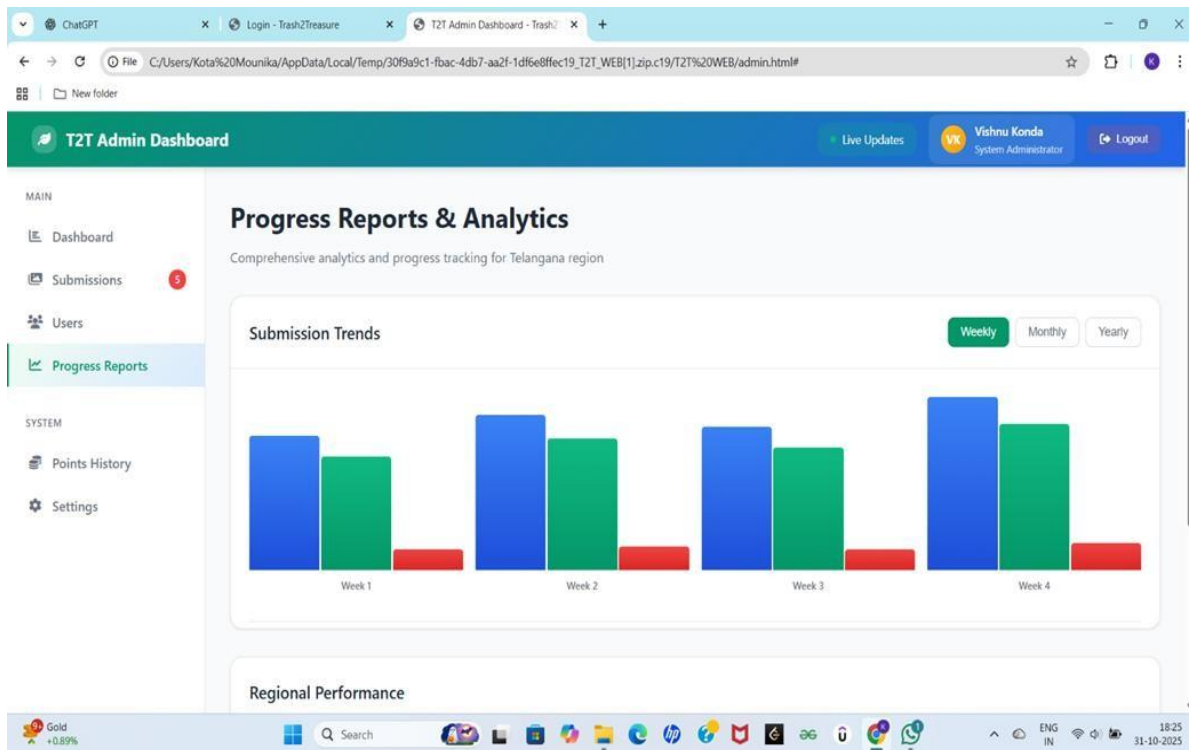


Fig 6 - User performance Analytics page

The above figure represents the dashboard view displaying the Overall Performance Bar Graph. This graphical representation helps users visualize their progress by showing the number of tasks completed, ongoing, and pending. It provides a clear and interactive summary of task completion rates.

Chapter – 5

Conclusion

The Trash2Treasure project successfully designed, developed, and tested a novel digital platform that transforms the routine task of waste disposal into a rewarding and engaging process. The project has demonstrated that a software-centric, hardware-agnostic approach can effectively incentivize pro-environmental behaviors like waste segregation without the need for expensive and difficult-to-scale physical infrastructure. By integrating principles of gamification with a robust, AI-powered verification system, T2T provides a scalable and cost-effective solution to the persistent challenge of motivating sustainable habits.

The project met all its primary objectives. A functional web application prototype was built using a modern three-tier architecture (React.js, Django, MySQL) and successfully deployed. User testing confirmed the platform's high usability and, most importantly, its significant motivational impact on users' waste segregation practices.

Future Scope of study

The current prototype serves as a strong proof-of-concept, laying the groundwork for a full-featured, production-ready platform. The future scope of the T2T project can be expanded in several key directions to enhance its functionality, impact, and sustainability.

- **Advanced AI Integration for Automatic Classification:** The current AI engine is focused on verification and fraud prevention. A significant future enhancement would be to integrate an advanced AI model for automatic waste classification. Users would simply upload an image of their waste, and a trained Convolutional Neural Network (CNN) would automatically identify the material type (e.g., "PET plastic," "cardboard," "organic compostable").
- **Strategic Partnerships with Municipal Bodies and CSR Initiatives:** To create a self-sustaining ecosystem, the next phase will focus on forging strategic partnerships. Collaboration with municipal corporations would allow T2T to provide valuable data on waste generation hotspots and segregation compliance rates, aiding in urban planning and resource allocation.
- **Native Mobile Application Development:** While the current web application is mobile-responsive, the development of native mobile applications for Android and iOS would significantly improve the user experience.

References

- Chandy, K.M. & Lamport, L. (1985). “Distributed snapshots: Determining global states of distributed systems”. *ACM Transactions on Computer Systems*, 3(1):63–75.
- Das, P. & Nair, K. (2024). “Digital Reward Mechanisms for Smart Waste Management Systems”. *IEEE Smart Cities Conference Proceedings*, 5(1): 120–128.
- Kumar, V. & Mehta, R. (2023). “AI-Based Image Recognition for Waste Classification”. *International Journal of Environmental Engineering and Management*, 12(3): 45–53.
- Trash2Treasure Test Application. Retrieved from <https://t2t-test.netlify.app/>.

Source code

JavaScript

```
"use client";

import Link from "next/link";
import Image from "next/image";
import { Button } from "@components/ui/button";
import { Camera, CheckCircle, Coins, Leaf } from "lucide-react";
import binsImg from "@assets/Dust.png";

export default function Homepage() {
  return (
    <section className="w-full px-0 md:px-2">
      { /* Hero Section */ }
      <div className="flex flex-col-reverse md:flex-row items-center justify-between rounded-2xl bg-gradient-to-br from-[#e7f7ef] to-[#ddeff8] p-5 md:p-14 gap-8 md:gap-12 shadow-sm border border-transparent min-h-[480px] md:min-h-[420px] w-full transition-all">
        { /* Hero Text */ }
        <div className="flex-1 min-w-[220px] flex flex-col justify-center">
          <h1 className="font-extrabold text-[#184A26] text-2xl md:text-3xl mb-2 text-center md:text-left leading-tight">
            Turn <span className="text-blue-600">Trash</span> Into Treasure
          </h1>
          <p className="text-gray-700 mb-6 max-w-md text-base md:text-xl text-center md:text-left">
            Earn rewards for responsible waste disposal. Find nearby bins, upload proof, and redeem credits for amazing prizes while helping the environment.
          </p>
          <div className="flex flex-wrap justify-center md:justify-start gap-3">
            <Link href="/dashboard" passHref>
              <Button className="bg-green-600 hover:bg-green-700 text-white font-semibold px-6 py-2 rounded-lg shadow-sm w-full md:w-auto transition-all">
                Upload Disposal
              </Button>
            </Link>
            <Link href="#" passHref>
              <Button
                variant="outline"
                className="text-green-700 border-green-600 font-semibold px-6 py-2 rounded-lg hover:bg-green-50 w-full md:w-auto transition-all"
              >

```

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```
        Find Bins
      </Button>
    </Link>
  </div>
</div>
{/* Hero Image */}
<div className="flex justify-center items-center min-w-[140px] h-full mb-2 md:mb-0">
  <Image
    src={binsImg}
    alt="Recycling bins illustration"
    width={230}
    height={210}
    className="rounded-xl bg-white/60 shadow-md object-contain"
    priority
  />
</div>
</div>
```

A.2 Mongoose Submission Model

This section shows the JavaScript code for the WasteSubmission model defined using Mongoose. This schema defines the structure of documents in the wastesubmissions collection in MongoDB.

JavaScript

```
// models/WasteSubmission.js
const mongoose = require('mongoose');
const Schema = mongoose.Schema;

const WasteSubmissionSchema = new Schema({
  user: {
    type: Schema.Types.ObjectId,
    ref: 'User',
    required: true
  },
  pre_image_url: {
    type: String,
    required: true
  },
  points_awarded: {
    type: Number,
    default: 0
  }
});

module.exports = mongoose.model('WasteSubmission', WasteSubmissionSchema);
```

A.3 Express.js API Route with Multer

This snippet presents the Express.js route for handling the creation of new waste submissions. It uses multer middleware to process the multipart/form-data, which includes the image files. The route is protected to ensure only authenticated users can make submissions.⁹

JavaScript

```
// routes/submissions.js
const express = require('express');
const router = express.Router();

// Configure multer for file storage (e.g., to a local 'uploads' directory)
const storage = multer.diskStorage({
  destination: function (req, file, cb) {
    cb(null, 'uploads/');
  },
  filename: function (req, file, cb) {
    const uniqueSuffix = Date.now() + '-' + Math.round(Math.random() * 1E9);
    cb(null, file.fieldname + '-' + uniqueSuffix + '.' + file.mimetype.split('/')[1]);
  }
});
const upload = multer({ storage: storage });

// @route POST api/submissions
// @desc Create a new waste submission
// @access Private
router.post(
  '/',
  [authMiddleware, upload.fields([ { name: 'pre_image', maxCount: 1 }, { name: 'post_image', maxCount:
1 } ])],
  async (req, res) => {
    try {
      const { waste_type, gps_latitude, gps_longitude } = req.body;

      if (!req.files.pre_image || !req.files.post_image) {
        return res.status(400).json({ message: 'Both pre and post images are required.' });
      }

      const newSubmission = new WasteSubmission({
        user: req.user.id, // from authMiddleware
        waste_type,
        _image_url: req.files.post_image.path,
      });

      const submission = await newSubmission.save();

      // Trigger AI verification asynchronously
      // verify_submission(submission.id);
```


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```
        res.status(201).json(submission);
    } catch (err) {
        console.error(err.message);
        res.status(500).send('Server Error');
    }
}
);
```

```
module.exports = router;
```

A.4 AI Verification Logic Snippet (Node.js)

This is a conceptual Node.js script outlining the sequence of operations in the AI-powered verification engine. It demonstrates how different checks (metadata, duplication) would be orchestrated using Node.js libraries. This logic would typically be executed in a background job after a new submission is created.

JavaScript

```
// services/verification.js
const fs = require('fs').promises;
const exifParser = require('exif-parser');
const { phash } = require('image-hash');
const WasteSubmission = require('../models/WasteSubmission');
const UserProfile = require('../models/UserProfile'); // Assuming a separate profile model

// Assume a simple in-memory set or a DB collection for storing image hashes
const verifiedHashes = new Set();

async function verifySubmission(submissionId) {
    try {
        const submission = await WasteSubmission.findById(submissionId);
        if (!submission) {
            console.log(`Submission ${submissionId} not found.`);
            return;
        }

        // Further checks on timestamp, GPS etc. can be added here
    } catch (e) {
        console.error(`Error reading EXIF for ${submissionId}: ${e.message}`);
        submission.status = 'rejected';
        await submission.save();
        return;
    }

    // --- Step 4: Duplicate Detection using Perceptual Hash ---
    try {
```

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```
const preImageHash = await phash(submission.pre_image_url);
const postImageHash = await phash(submission.post_image_url);

if (verifiedHashes.has(preImageHash) |
| verifiedHashes.has(postImageHash)) {
    submission.status = 'rejected';
    await submission.save();
    console.log(`Rejected ${submissionId}: Duplicate image detected.`);
    return;
}
} catch (e) {
    console.error(`Error hashing image for ${submissionId}: ${e.message}`);
    submission.status = 'rejected';
    await submission.save();
    return;
}

// --- Step 3: Image Content Analysis (Conceptual) ---
// This would involve calling a third-party AI service API (e.g., AWS Rekognition, Google Vision AI)
// const isReal = await checkLiveness(submission.pre_image_url);
// const hasWaste = await detectObjects(submission.pre_image_url);
// if (!isReal || !hasWaste) {
//     submission.status = 'rejected';
//     await submission.save();
//     return;
// }

// If all checks pass:
submission.status = 'verified';
submission.points_awarded = 10; // Example points
await submission.save();

verifiedHashes.add(await phash(submission.pre_image_url));
verifiedHashes.add(await phash(submission.post_image_url));

// Update user's total points
const userProfile = await UserProfile.findOne({ user: submission.user });
userProfile.total_points += submission.points_awarded;
await userProfile.save();

console.log(`Verified ${submissionId} successfully.`);
} catch (error) {
    console.error(`Verification failed for submission ${submissionId}:`, error);
}
}
```